

If your goal is **deep stock-price prediction research** and **future-facing quantitative AI**, then I would remove:

- Basic package documentation
- Introductory Python resources
- Beginner ML courses
- Generic tutorials

Instead, focus on **research labs, advanced frameworks, quantitative research platforms, papers-with-code ecosystems, forecasting foundations, transformers, probabilistic forecasting, and financial machine learning.**

Top 20 Advanced Resources for Stock Price Prediction Research

1. Machine Learning for Trading (Stefan Jansen)

Perhaps the most complete open-source quantitative research resource.

Focus:

- Alpha research
- Feature engineering
- Alternative data
- Deep learning
- Portfolio construction

🔗 <https://github.com/stefan-jansen/machine-learning-for-trading>

2. ML4Trading

Advanced financial machine learning platform inspired by professional quant workflows.

Topics:

- Financial ML
- Backtesting
- Alpha Factors
- Research Pipelines

🔗 [ML4Trading](#)

3. Hudson & Thames

One of the best quantitative-research organizations.

Topics:

- Financial Machine Learning
- Portfolio Optimization
- Feature Engineering
- Market Microstructure

🔗 [Hudson & Thames](#)

4. Marcos López de Prado Research

The intellectual foundation of modern financial ML.

Study:

- Triple Barrier Labeling
- Meta Labeling
- Fractional Differentiation
- Financial Cross Validation

🔗 <https://www.quantresearch.org>

5. Quant Research Papers Repository

Large collection of quantitative-finance papers.

🔗 [Quantocracy](#)

Research Areas:

- Alpha Factors
- Factor Investing
- Market Prediction
- Deep Learning Finance

Time Series Foundation Models

These represent the future of stock prediction.

6. Google TimesFM

Google's foundational time-series forecasting model.

Topics:

- Zero-shot forecasting
- Large-scale temporal modeling

- Foundation models

🔗 <https://github.com/google-research/timesfm>

7. Amazon Chronos

One of the most important forecasting foundation models.

Research:

- Transformer forecasting
- Time-series tokenization

🔗 <https://github.com/amazon-science/chronos-forecasting>

8. Salesforce Moirai

Next-generation universal forecasting model.

Research Topics:

- Cross-domain forecasting
- Large-scale pretrained models

🔗 <https://github.com/SalesforceAIResearch/uni2ts>

9. Nixtla TimeGPT

The closest equivalent of "ChatGPT for forecasting."

Research:

- Foundation forecasting
- Financial forecasting
- Zero-shot forecasting

🔗 <https://www.nixtla.io>

10. Lag-Llama

Modern large-scale forecasting architecture.

Topics:

- Probabilistic forecasting
- Multi-series prediction

🔗 <https://github.com/time-series-foundation-models/lag-llama>

Advanced Time-Series Deep Learning

11. Darts

Possibly the most comprehensive forecasting framework available.

Includes:

- TFT
- N-BEATS
- Transformer
- N-HiTS
- TCN
- TiDE

🔗 <https://unit8co.github.io/darts>

12. NeuralForecast

Professional forecasting framework.

Includes:

- PatchTST
- NBEATS
- TFT
- NHITS
- TimesNet

🔗 <https://github.com/Nixtla/neuralforecast>

13. PyTorch Forecasting

Even if you primarily use TensorFlow, this repository contains state-of-the-art forecasting implementations.

🔗 [PyTorch Forecasting](#)

Focus:

- Temporal Fusion Transformer
- DeepAR
- Attention mechanisms

14. Time Series Library (TSLib)

One of the most important benchmark repositories.

Models:

- Informer
- Autoformer
- FEDformer
- TimesNet
- PatchTST

🔗 <https://github.com/thuml/Time-Series-Library>

Research Papers Ecosystem

15. Papers With Code - Time Series Forecasting

Track current state-of-the-art models.

🔗 [Papers with Code Forecasting](#)

Research:

- Benchmark rankings
- New architectures
- Source code

16. arXiv Quantitative Finance

Primary source of cutting-edge research.

🔗 [arXiv Quantitative Finance](#)

Monitor:

- Market forecasting
- Deep learning finance
- Reinforcement learning trading

17. arXiv Time Series Papers

Future forecasting research appears here first.

🔗 [arXiv Time Series](#)

Search Topics:

- TimesNet
- PatchTST
- TimeMixer

- Time-MoE

Institutional-Grade Quant Research

18. QuantConnect Research

Industry-level quantitative research platform.

Topics:

- Alpha factors
- Market research
- Walk-forward testing

🔗 [QuantConnect Research](#)

19. VectorBT

Modern vectorized research and backtesting ecosystem.

Ideal for:

- Feature testing
- Strategy research
- Massive parameter sweeps

🔗 [VectorBT](#)

20. PyPortfolioOpt

Critical for turning forecasts into portfolio decisions.

Focus:

- Risk models
- Portfolio optimization
- Efficient frontier

🔗 [PyPortfolioOpt](#)

Future-Focused Learning Order (2026-2030)

If your objective is becoming a serious AI-driven quantitative researcher, I recommend this progression:

1 Phase 1

```
2 Financial ML
3   ├── ML4Trading
4   ├── Hudson & Thames
5   └── López de Prado
6
7 Phase 2
8 Research Infrastructure
9   ├── VectorBT
10  ├── QuantConnect
11  └── PyPortfolioOpt
12
13 Phase 3
14 Advanced Forecasting
15   ├── Darts
16   ├── NeuralForecast
17   └── TSLib
18
19 Phase 4
20 Transformer Forecasting
21   ├── PatchTST
22   ├── TimesNet
23   ├── TFT
24   └── Informer
25
26 Phase 5
27 Foundation Models
28   ├── TimesFM
29   ├── Chronos
30   ├── Moirai
31   ├── TimeGPT
32   └── Lag-Llama
33
34 Phase 6
35 Research Frontier
36   ├── Papers With Code
37   ├── arXiv
38   └── New Forecast Foundation Models
```

My "Top 5 Highest ROI" resources for the next few years

1. **ML4Trading**
2. **Hudson & Thames**

3. **NeuralForecast**
4. **Google TimesFM**
5. **TSLib (Time-Series-Library)**

Together, those five resources cover classical quantitative finance, modern machine learning, transformer forecasting, foundation models, and the current research frontier for stock-price prediction.